

DATA 260 Homework 2 Report

Student and Configuration

- Student: Vrishin Dharmesh Kunnatham Parambath
- SID4: 8844
- PORT_BASE: 8744
- PREFIX: s8844
- SEED: 8844
- VERIFY_SEED: 268844
- DOMAIN_ID: 4
- Assigned domain: Open-source package vulnerabilities
- Model: qwen3:8b
- Hardware: Windows 11 laptop
- Final tag: hw2
- GitHub repository: <https://github.com/Viki2183/data260-8844>

Part 1 — HTML and CSS

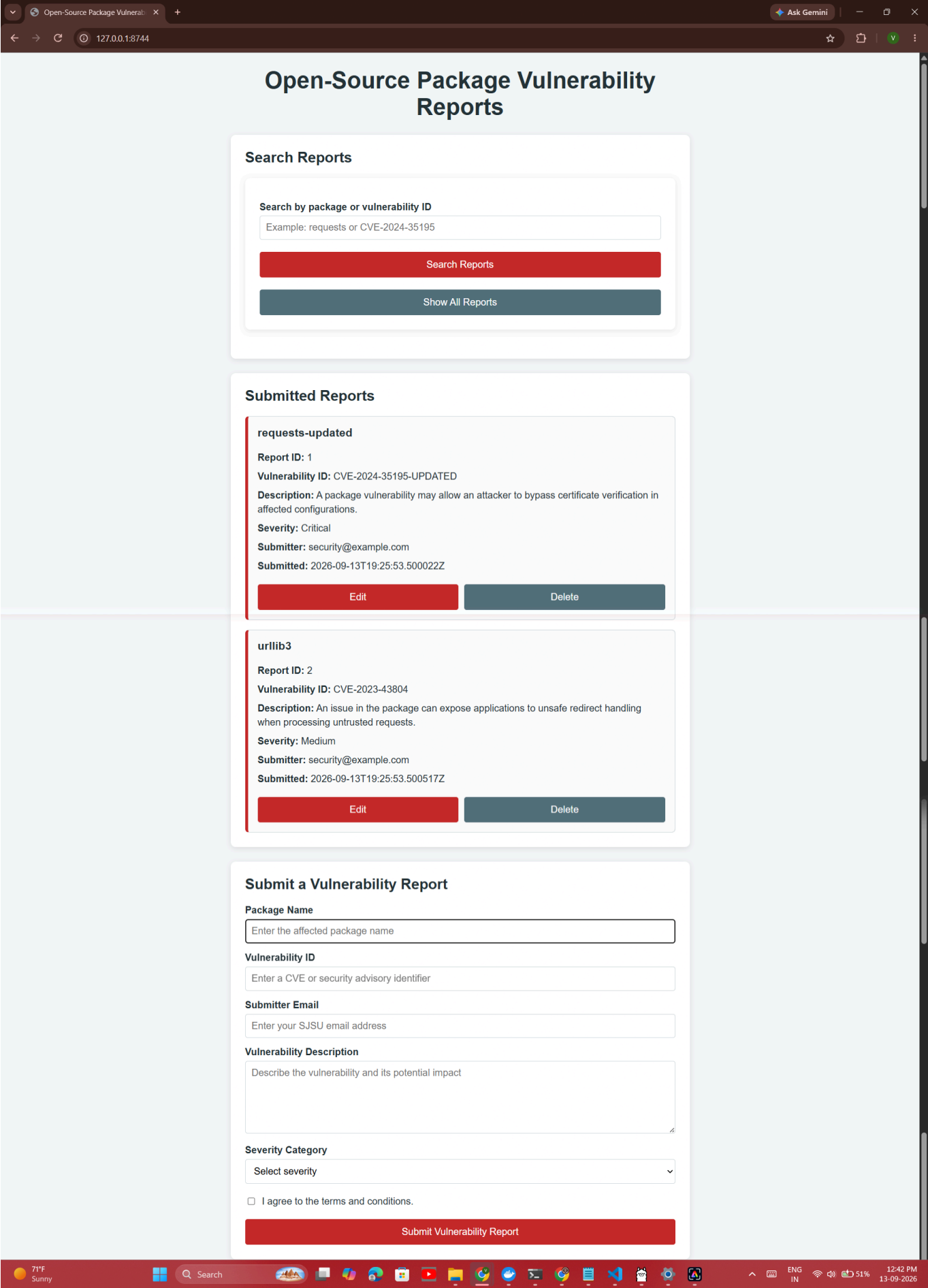
The HW1 vulnerability form was extended with a responsive layout. The page uses an external `styles.css` file and includes a mobile media query for narrow screens, including a 375px-wide display.

The interface includes:

- A vulnerability report submission form
- A search form
- Report cards
- Edit and delete buttons
- A visible loading state
- A visible empty state
- A visible error state

The page was tested through the Dockerized application at:

<http://127.0.0.1:8744>



Open-Source Package Vulnerability Reports

Search Reports

Search by package or vulnerability ID

Example: requests or CVE-2024-35195

Search Reports

Show All Reports

Submitted Reports

requests-updated

Report ID: 1

Vulnerability ID: CVE-2024-35195-UPDATED

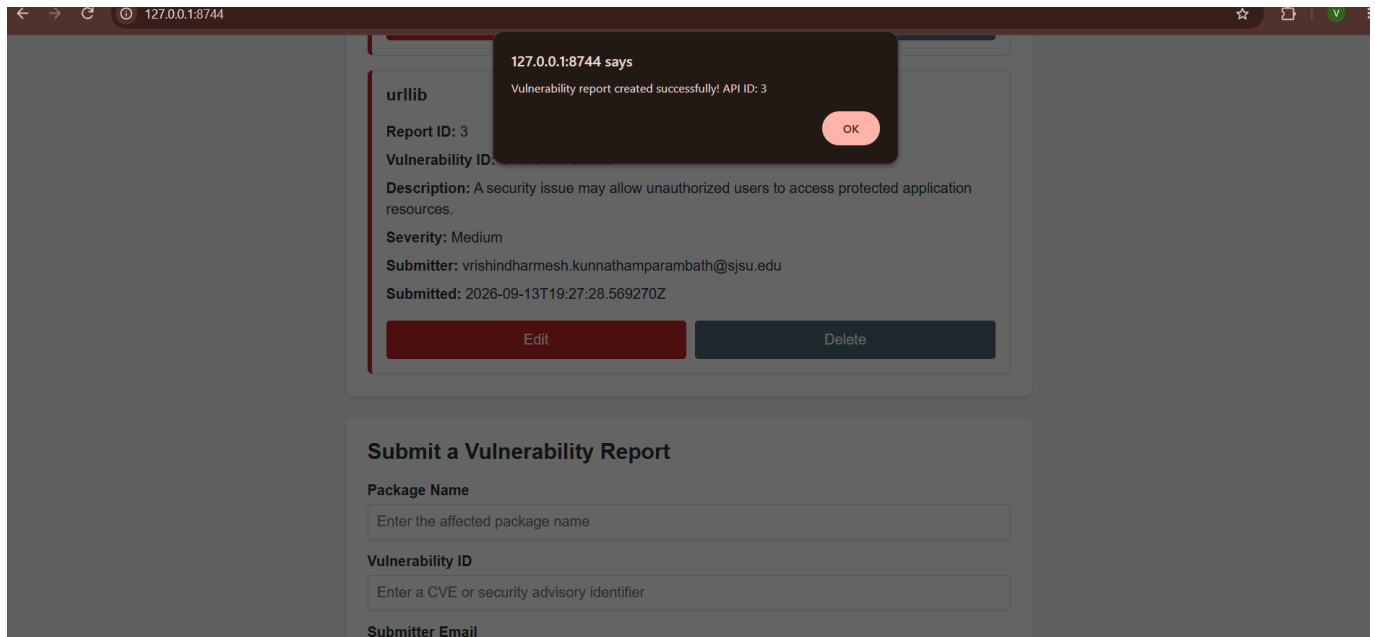
Description: A package vulnerability may allow an attacker to bypass certificate verification in affected

Part 2 — FastAPI

The FastAPI backend runs on `PORT_BASE` 8744 and provides REST endpoints for vulnerability reports.

Part 2.1 — Create a Record

The frontend sends a POST request to `/api/vulnerability-reports`. A new report was created successfully and returned API ID 3.



Part 2.2 — Update Record ID 1

The report with ID 1 was edited through the webpage. Its package name was changed to `requests-updated`, its vulnerability ID was changed to `CVE-2024-35195-UPDATED`, and its severity was changed to `Critical`.

Edit Report 1

Package Name

requests-updated

Vulnerability ID

CVE-2024-35195-UPDATED

Submitter Email

security@example.com

Vulnerability Description

A package vulnerability may allow an attacker to bypass certificate verification in affected configurations.

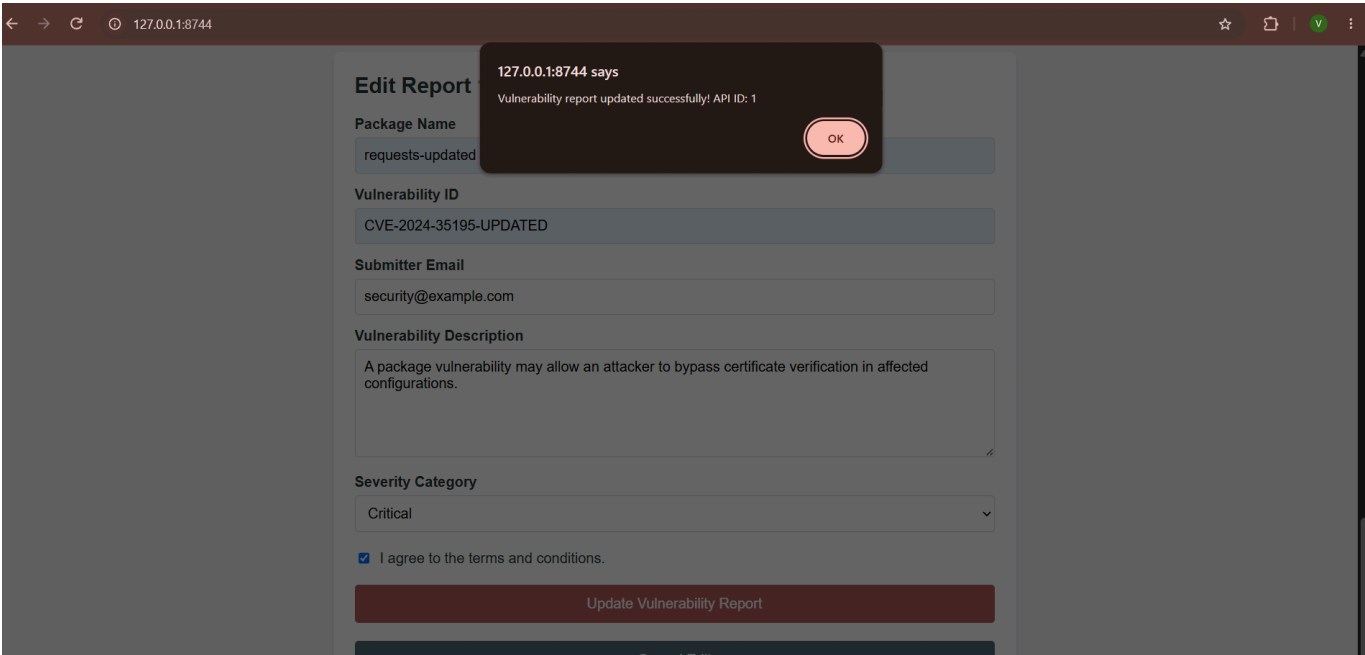
Severity Category

Critical

☒ I agree to the terms and conditions.

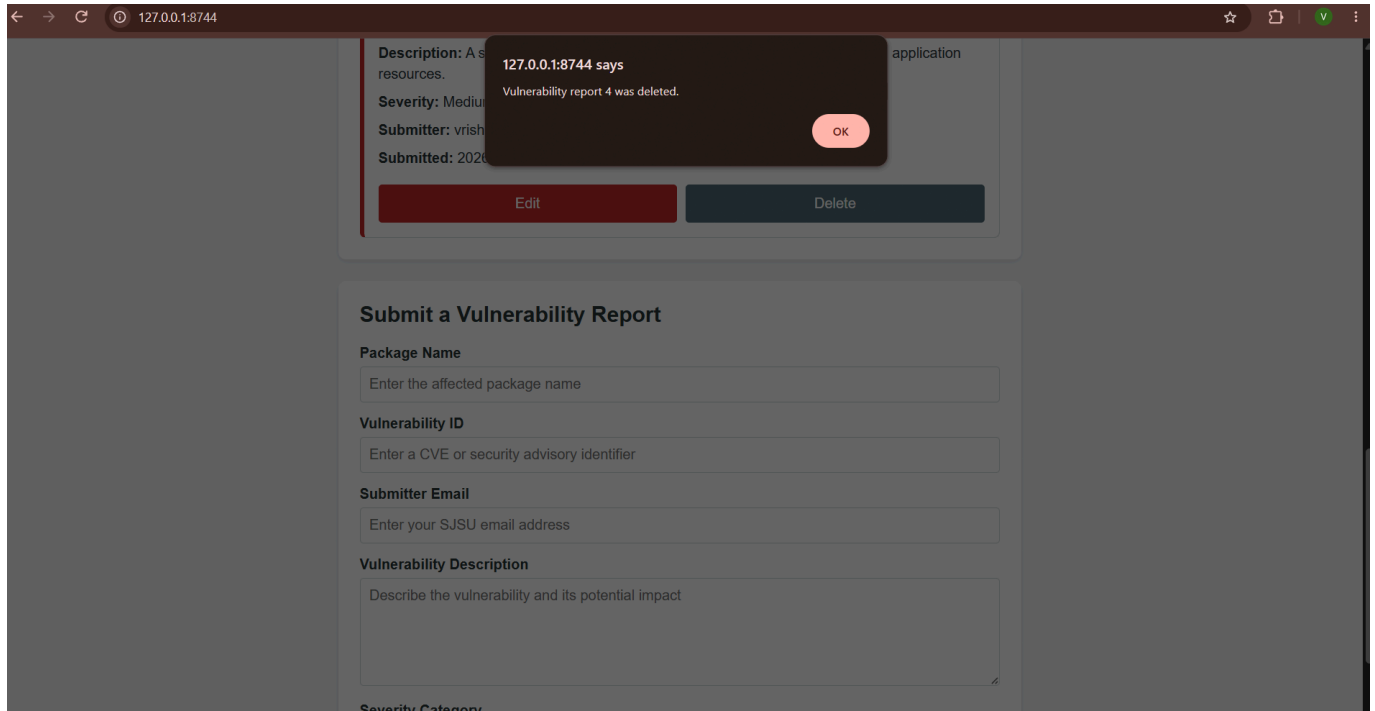
Update Vulnerability Report

Cancel Edit



Part 2.3 — Delete the Highest-ID Record

The temporary highest-ID report, Report ID 4, was deleted through the webpage. The success message confirmed the deletion, and the refreshed report list showed that Report ID 4 was no longer present.



Open-Source Package Vulnerability Reports

Search Reports

Search by package or vulnerability ID

Example: requests or CVE-2024-35195

Search Reports

Show All Reports

Submitted Reports

requests-updated

Report ID: 1

Vulnerability ID: CVE-2024-35195-UPDATED

Description: A package vulnerability may allow an attacker to bypass certificate verification in affected configurations.

Severity: Critical

Submitter: security@example.com

Submitted: 2026-09-13T19:25:53.500022Z

EditDelete

urllib3

Report ID: 2

Vulnerability ID: CVE-2023-43804

Description: An issue in the package can expose applications to unsafe redirect handling when processing untrusted requests.

Severity: Medium

Submitter: security@example.com

Submitted: 2026-09-13T19:25:53.500517Z

EditDelete

Submit a Vulnerability Report

Package Name

Enter the affected package name

Vulnerability ID

Enter a CVE or security advisory identifier

Submitter Email

Enter your SJSU email address

Vulnerability Description

Describe the vulnerability and its potential impact

Severity Category

Select severity

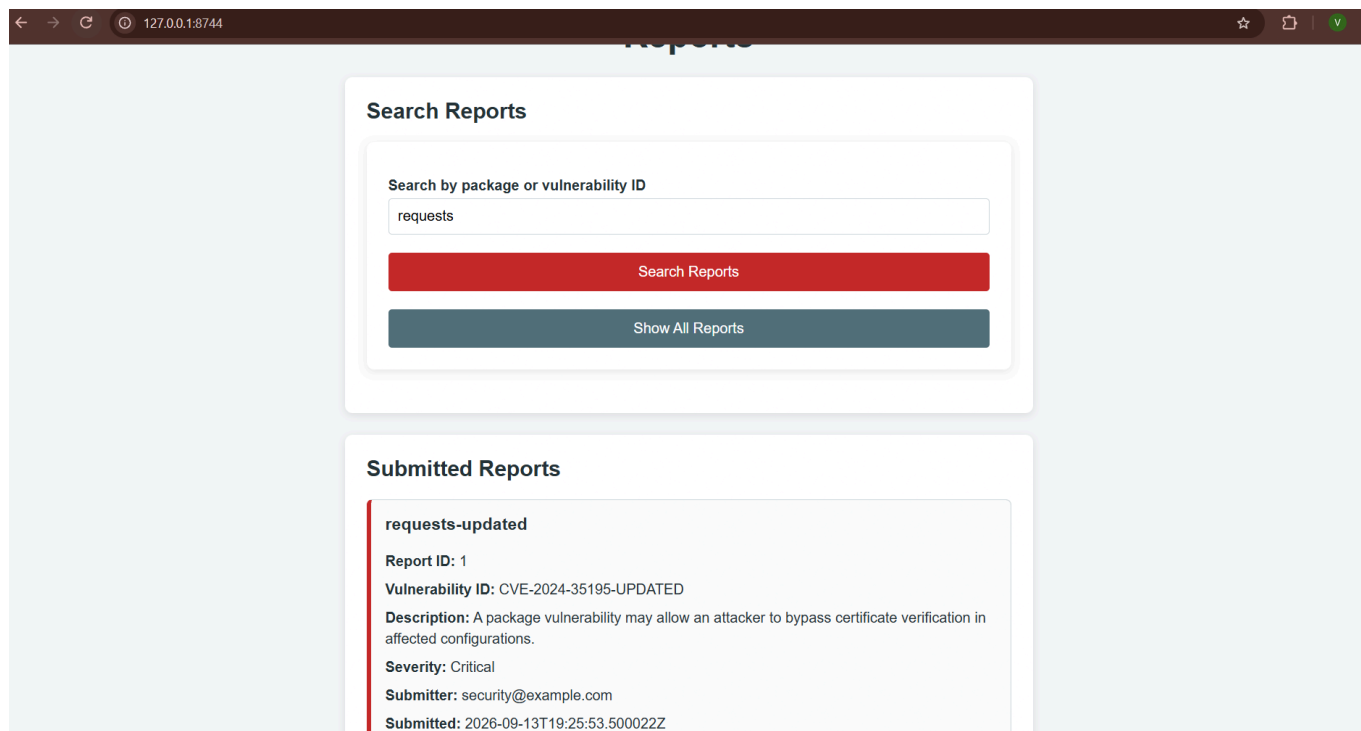
☐ I agree to the terms and conditions.

Submit Vulnerability Report

Part 2.4 — Search

Searching for **requests** returned only the matching report.

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Part 3 — Stateful Agent Graph

The sequential HW1 workflow was extended into a stateful LangGraph workflow in [code/hw2_graph.py](#).

All LLM calls in [hw2_graph.py](#), including the Planner and Reviewer nodes, are routed through the shared HW1 adapter in [src/model_client.py](#). This preserves the common Ollama configuration and token-accounting behavior.

The graph contains:

- [AgentState](#), which stores shared workflow memory
- A Planner node
- A Reviewer node
- A Supervisor node
- Conditional routing logic
- A turn counter to prevent infinite loops
- A retry path when validation or review fails

The Planner output is validated with Pydantic. The schema requires exactly three distinct string tags, each between 3 and 30 characters, and a summary containing no more than 25 words.

The graph was tested with the local [qwen3:8b](#) model. The test completed in 3 supervisor turns, and the Reviewer approved the output.

The final output contained exactly three tags:

- [AuthenticationBypass](#)
- [RequestsPackage](#)
- [UnauthorizedAccess](#)

The generated summary was within the 25-word limit.

```

C:\Projects\data260-SB44\venv\Lib\site-packages\langgraph\cache\base\_init_.py:8: LangChainPendingDeprecationWarning: The default value of 'allowed_objects' will change in a future version. Pass an explicit value (e.g., allowed_objects='messages' or allowed_objects='core') to suppress this warning.
  from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer
-- NODE: Supervisor (turn 1/10) ---
-- NODE: Planner ---
turn token usage: input=160, output=258, total=418
Lanner produced valid output in 76572 ms.
-- NODE: Supervisor (turn 2/10) ---
-- NODE: Reviewer ---
turn token usage: input=160, output=344, total=513

{
  "approved": true,
  "issues": [],
  "latency_ms": 82083.95
}
-- NODE: Supervisor (turn 3/10) ---
-- GRAPH RESULT ---
{
  "title": "Critical Authentication Bypass in Requests Package",
  "email": "vishindharmesh.kunnathparamath@rsu.edu",
  "model": "gpt4o",
  "temperature": 0.8,
  "max_turns": 10,
  "turn_count": 3,
  "completed": true,
  "final_output": {
    "tags": [
      "AuthenticationBypass",
      "RequestsPackage",
      "UnauthorizedAccess"
    ],
    "summary": "Critical authentication bypass allows unauthorized access to protected resources and sensitive data via the Requests package."
  },
  "reviewer_feedback": {
    "approved": true,
    "issues": [],
    "latency_ms": 82083.95
  },
  "validation_error": "",
  "token_stats": {
    "turn_count": 2,
    "cumulative_input_tokens": 320,
    "cumulative_output_tokens": 602,
    "cumulative_total_tokens": 922,
    "serialized_history_length": 2
  }
}
C:\Projects\data260-SB44> |

```

Part 4 — Schema Validation and Loop Safety

Schema Validation Experiment

The frozen input was stored in:

reports/hw02/cases/schema_input.json

The input was run 30 times.

Outcome	Count
Valid first attempt	30
Valid after 1 retry	0
Valid after 2 or more retries	0
Hit turn ceiling	0

Completion rate: 100%

Mean latency: 178,399.88 ms

Turn Ceiling Comparison

Each ceiling was tested 20 times using the same frozen input and model settings.

Turn ceiling	Runs	Completed	Completion rate	Mean latency
2	20	0	0%	71,869.60 ms
10	20	20	100%	157,746.71 ms

A ceiling of 2 was too small because the workflow normally requires Supervisor, Planner, and Reviewer turns before it can finish.

A ceiling of 10 completed all runs successfully, so ceiling 10 was selected for deployment.

Adversarial Experiment

The adversarial input was stored in:

`reports/hw02/cases/adversarial_input.json`

It was run five times.

Outcome	Count
Completed	5
Hit turn ceiling	0

Completion rate: 100%

Observed ceiling rate: 0%

Mean latency: 134,891.41 ms

The adversarial input was designed to create difficulty by combining ambiguous wording, conflicting instructions, and content that could cause the Planner or Reviewer to produce output that violates the required schema. Such input could lead to invalid tags, an overly long summary, repeated correction attempts, or a turn-ceiling failure.

In the five observed runs, the graph completed successfully every time and did not reach the turn ceiling. This observed result is reported honestly rather than claiming that the input reliably causes failure. One improvement would be to add stronger schema-aware retry feedback that explicitly identifies the invalid field and requires the next model response to correct only that field. A separate maximum retry limit would also prevent repeated correction loops.

```

PS C:\Projects\data260-8844> Get-Content ".\reports\hw02\raw\experiment_summary.json"
{
  "schema_validation": {
    "run_count": 30,
    "completion_count": 30,
    "completion_rate": 1.0,
    "mean_latency_ms": 178399.88,
    "classifications": {
      "Valid first attempt": 30
    }
  },
  "ceiling_2": {
    "run_count": 20,
    "completion_count": 0,
    "completion_rate": 0.0,
    "mean_latency_ms": 71869.6,
    "classifications": {
      "Hit turn ceiling": 20
    }
  },
  "ceiling_10": {
    "run_count": 20,
    "completion_count": 20,
    "completion_rate": 1.0,
    "mean_latency_ms": 157746.71,
    "classifications": {
      "Valid first attempt": 20
    }
  },
  "adversarial": {
    "run_count": 5,
    "completion_count": 5,
    "completion_rate": 1.0,
    "mean_latency_ms": 134891.41,
    "classifications": {
      "Valid first attempt": 5
    }
  }
}
PS C:\Projects\data260-8844> |

```

Docker Deployment

The FastAPI application was packaged in a Docker image named:

`data260-8844-hw2`

The container was started with:

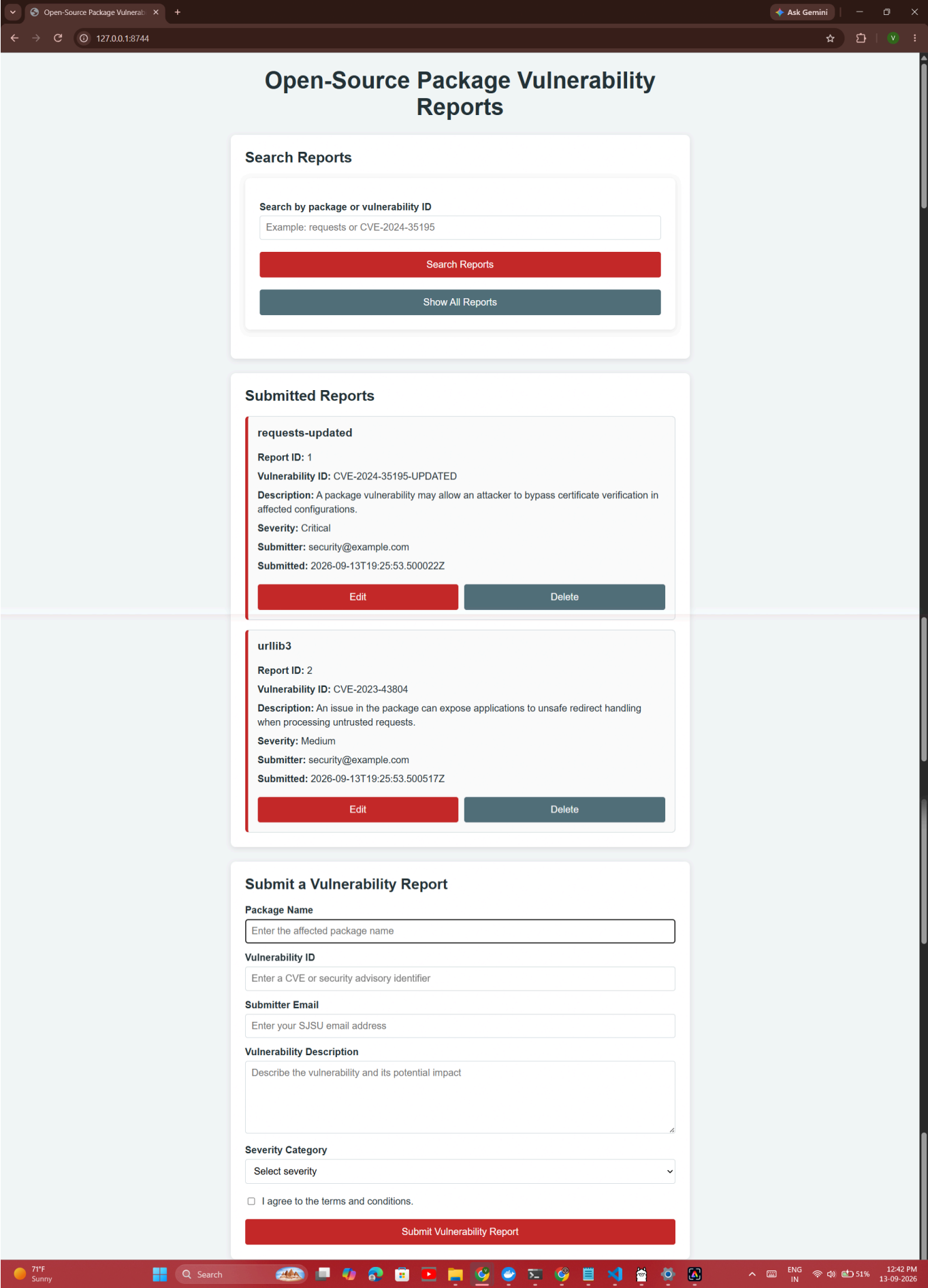
```
docker run -d -p 8744:80 --name data260-8844-hw2-container data260-8844-hw2
```

The container started successfully, and the API responded with HTTP 200. The webpage also loaded successfully from the Docker container.

```

PS C:\Projects\data260-8844> docker ps --filter "name=data260-8844-hw2-container"
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
679d003d618d   data260-8844-hw2   "uvicorn web_applica..."  14 hours ago   Up 15 minutes   0.0.0.0:8744->80/tcp, [::]:8744->80/tcp   data260-8844-hw2-container
PS C:\Projects\data260-8844> |

```



Verification

The verification script is:

`code/verify_hw02.py`

It checks:

- Required HW2 files
- FastAPI API response
- LangGraph smoke-test completion

The final verification produced:

```
{
  "all_passed": true
}
```

```
PS C:\Projects\data260-8844> Get-Content ".\reports\hw02\verification.json"
{
  "homework": "DATA 260 Homework 2",
  "sid4": 8844,
  "port_base": 8744,
  "seed": 8844,
  "verify_seed": 268844,
  "model": "qwen3:8b",
  "timestamp_utc": "2026-09-13T18:59:55.365903+00:00",
  "commit_hash": "e09e8dcfcb2c657a0c20b1019258b2f50b38c9ee",
  "checks": [
    {
      "name": "Required HW2 files",
      "passed": true,
      "details": "All required files exist."
    },
    {
      "name": "FastAPI API response",
      "passed": true,
      "details": "HTTP 200; received 3 reports"
    },
    {
      "name": "LangGraph smoke test",
      "passed": true,
      "details": "Return code 0; GRAPH RESULT found"
    }
  ],
  "all_passed": true
}
PS C:\Projects\data260-8844> |
```

Reproducibility

The following files contain the reproducible HW2 evidence:

- `reports/hw02/RUN_LOG.txt`

- reports/hw02/raw/schema_validation_results.json
- reports/hw02/raw/schema_validation_results.csv
- reports/hw02/raw/ceiling_comparison_results.json
- reports/hw02/raw/ceiling_comparison_results.csv
- reports/hw02/raw/adversarial_results.json
- reports/hw02/raw/experiment_summary.json
- reports/hw02/verification.json

The main commands are:

```
python .\code\run_hw02_experiments.py
python .\code\verify_hw02.py
docker build -f .\code\Dockerfile -t data260-8844-hw2 .
docker run -d -p 8744:80 --name data260-8844-hw2-container data260-8844-hw2
```

Conclusion

This homework extended the HW1 vulnerability-report application into a Dockerized FastAPI application with CRUD operations, search, responsive styling, visible interface states, and a stateful LangGraph agent workflow. The measured experiments showed that a turn ceiling of 10 was reliable for the tested input, while a ceiling of 2 was insufficient.

- Final tag: hw2
- Tagged commit hash: a3fcc78ceb5efc75a7d38041403365837d7f25b3